

Discussion and further work

Although an LTU Attacker is all knowledgeable, it must make efficient use of available information to be powerful. We proposed a taxonomy based on information available or used (Figure 2). The most powerful Attackers use both the trained Defender model $\mathcal{M}_D$ and its trainer $\mathcal{T}_D$.

When the Defender trainer $\mathcal{T}_D$ is a black box, like in our first set of experiments on scikit-learn algorithms, we see clear limitations of the LTU Attacker which include the fact that it is not possible to diagnose whether the algorithm is example-based.

Unfortunately, white-box attacks cannot be conducted in a generic way, but must be tailored to the trainer (*e.g.*, gradient descent algorithms for MLP). In contrast, black-box methods can attack $\mathcal{T}_D$ (and $\mathcal{M}_D$) regardless of mechanism. Still, we get necessary conditions for privacy protection by analyzing black-box methods. Both theoretical and empirical results using black-box attackers (on a broad range of algorithms of the scikit-learn library on the QMNIST and CIFAR-10 data), indicate that Defender algorithms are vulnerable to a LTU Attacker if it overfits the training Defender data or if it is deterministic. Additionally, the degree of stochasticity of the algorithm must be sufficient to obtain a desired level of privacy.

We explored white-box attacks neural networks trained with gradient descent. In our experiments on the large QM-NIST dataset (200,000 training examples), Deep CNNs such as ResNet seem to exhibit both good Utility and Privacy in their “native form”, according to our white-box attacker. We were pleasantly surprised of our white box attack results, but, in light of the fact that other authors found similar networks vulnerable to attack (?), we conducted the following sanity check. We performed the same supervised learning experiment by modifying 20% of the class labels (to another class label chosen randomly), in both the Defender set and Reserved set. Then we incited the neural network to overfit the Defender set. Although the training accuracy (on Defender data) was still nearly perfect, we obtained a loss of test accuracy (on Reserved data): 78%. According Theorem 2, this should result in a loss of privacy. This allowed us to verify that our white-box attacker correctly detected a loss of privacy. Indeed, we obtained a privacy of 0.55.

We are in the process of conducting comparison experiments between our white-box attacker and that of (?). However, their method does not easily lend itself to be used with the LTU framework, because it requires training a neural network for each LTU round (*i.e.*, on each $\mathcal{D}_A = \mathcal{D}_D - \{\text{membership}(d)\} \cup \mathcal{D}_R - \{\text{membership}(r)\}$). We are considering doing only one data split to evaluate privacy, with $\mathcal{D}_A = 50\% \mathcal{D}_D \cup 50\% \mathcal{D}_R$ and using the rest of the data for privacy evaluation. However, we can still use the pairwise testing of the LTU methodology, *i.e.*, the evaluator queries the attacker with pairs of samples, one from the Defender data and the other from the Reserved data. In Appendix C, we show on an example that this results in an increased accuracy of the attacker.

In Appendix C, we use the same example to illustrate how we can visualize the privacy protection of individuals. Further work includes comparing this approach with (?).

Further work also includes testing LTU Attacker on a wider variety of datasets and algorithms, varying the number of training examples, training new white-box attack variants to possibly increase the power of the attacker, and testing various means of improving the robustness of algorithms against attacks by LTU Attacker. We are also in the process of designing a competition of membership inference attacks.

Conclusion

In summary, we presented an apparatus for evaluating the robustness of machine learning models (Defenders) against membership inference attack, involving an “all knowledgeable” LTU Attacker. This attacker has access to the trained model of the Defender, its learning algorithm (trainer), all the Defender data used for training, *minus the label of one sample*, and all the *similarly distributed* non-training Reserved data (used for evaluation), *minus the label of one sample*. The Evaluator repeats this Leave-Two-Unlabeled (LTU) procedure for many sample pairs, to compute the efficacy of the Attacker, whose charter is to predict the membership of the unlabeled samples (training or non-training data). We call such LTU Attacker the LTU-attacker for short. The LTU framework helped us analyse privacy vulnerabilities both theoretically and experimentally.

The main conclusions of this paper are that a number of conditions are necessary for a Defender to protect privacy:

- Avoid storing examples (a weakness of example-based method, such as Nearest Neighbors).
- Ensure that $p_R = p_D$ for all f , following Theorem 1 (p_R is the probability that discriminant function f “favors” Reserved data while p_D is the probability with which it favors the Defender data).
- Ensure that $e_R = e_D$, following Theorem 2 (e_R is the expected value of the loss on Reserved data and e_D on Defender data).
- Include some randomness in the Defender trainer algorithm, after Theorem 3.

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